

PEDI PRODUCT SPECIFICATION BLUEPRINT

Competitive Feature Mapping: Xiaohongshu (RED) vs. Pedi Super-App Strategy

Target Market:	East Africa (Initial launch: Kenya; Expansion: Pan-African)
Core Stack:	Flutter (Mobile Frontend) + Firebase/Firestore (Real-time Cloud Backend)
Strategic Intent:	The absolute lifestyle, social commerce, and discovery operating system for African consumers.

This document outlines the formal technical and functional specification matrix for **Pedi**. Pedi adapts the highly successful closed-loop social commerce mechanics pioneered globally by Xiaohongshu (Little Red Book) and localizes them for the African market via short-form video discovery loops, frictionless localized payment pipelines, and an omnipresent conversational AI layer.

1. Global Feature Set of Xiaohongshu (RED)

- **'Notes' (Biji) Media Architecture:** Multi-image carousel uploads (up to 9 frames) anchored by dense, highly informative text captions functioning as structured lifestyle reviews rather than basic status updates.
- **Double-Column 'Explore' Feed:** A Pinterest-style staggered dual grid acting as the primary layout to drive active intent select-through rates and deep product discovery.
- **Short-Form Video Engine:** Full-screen vertical swipe-based media component dedicated to rapid ingestion of styling guides, try-on vlogs, and raw consumer unboxings.
- **'Grass Planting' (Zhong Cao) Bookmarking:** Psychological bookmarking loop enabling users to categorize physical spots, wishlists, and shopping reviews into public or private shared boards.
- **Geographic 'Nearby' Discovery:** Location telemetry mapping creator media posts to physical brick-and-mortar boutique stalls, local dining options, and local pop-up experiences.
- **Deep Search Intent Optimization:** Contextual text parsing using multi-tag recommendation chips, autocorrect loops, and filters ('Most Liked', 'Latest') optimized for intentional purchase research.
- **Clickable Product & Brand Tags:** Hotspot tags overlaid on media assets linking users directly to product configuration panels, pricing info, and native store links.
- **In-App Storefront Profiles:** Native vendor spaces hosting structured product collections, discount cards, and standalone brand pages.
- **Social Livestream Shopping:** Live streaming layout offering real-time video broadcasting embedded with an active checkout cart overlay and timed coupon releases.
- **Omnichannel Payment Vault:** Secure native single-tap payment vaults binding user accounts directly to WeChat Pay, Alipay, and international banking networks.
- **Direct-to-Chat Merchant Links:** Immediate secure instant messaging gateways connecting consumers directly to shop operations straight from the product asset card.
- **Enterprise Analytics Suite:** Comprehensive business dashboard tracking conversion funnels, keyword search velocities, retention graphs, and customer source metrics.

- **AI Search Assistant ('Diandian'):** Contextual large language model integration parsing user query intents to synthesize direct advice across millions of unstructured user reviews.

2. Localized Feature Set of Pedi (The African Super-App)

A. The Content & Local Discovery Layer

- **TikTok-Style Immersive Video UI:** Fluid 60fps vertical swipe architecture implemented via Flutter's `PageView.builder`, engineered specifically for immediate impulse discovery and engagement.
- **Double-Column Switch Matrix:** Home interface component enabling immediate switching between full-screen video streaming and a Pinterest-style dual-column grid optimized for targeted visual scanning.
- **Geo-Fenced Guide Engine:** Real-time location context using Firestore Geo-hashes and Flutter's geolocator telemetry. The discovery feed dynamically filters to display regional thrift plugs, hidden stays, and local spots relative to the user's location (Nairobi, Mombasa, Diani, Naivasha).
- **Contextual Infrastructure Plug ('Pedi wa Boda'):** Standardized metadata entries within travel and shopping vlogs allowing creators to tag precise logistics info (e.g., Matatu stages, expected fare bounds, and direct telephone hooks for verified local Boda Boda riders).
- **Offline Asset Persistence:** Native Firestore local storage persistence enabling local or international travelers to access their saved itineraries, infobox specs, and contact files completely offline in areas with zero cell coverage.

B. The Structured E-Commerce & Trust Infrastructure

- **The 'Wiki-Style' Product Infobox:** Standardized structural inputs completely removing the 'DM for price' friction. Enforces explicit fields for exact stall locations (e.g., *Imenti House, 2nd Floor, Stall 42*), fixed item prices, active inventory counts, and explicit regional delivery rates.
- **Interactive Hotspot Tagging (The 'VideoTag' Component):** Flutter UI engine rendering interactive tags on moving video panels. Tapping triggers a rapid `showModalBottomSheet` loading full Wiki specifications overlaying the current video window.
- **Typo-Tolerant Typographic Indexing:** Cloud Firestore collections synced via backend nodes to Algolia/Meilisearch instances to handle immediate predictive auto-complete and multi-tag filtering.

C. Frictionless Financial & Transaction Mechanics

- **Native M-Pesa STK Push Integration:** Firebase Cloud Functions securely bridging client checkouts directly to the Safaricom Daraja API. Pressing 'Buy Now' fires an instantaneous, secure M-Pesa SIM PIN prompt directly onto the user's viewport.
- **Multi-Channel Regional Gateway:** Native bindings via Paystack/Flutterwave APIs built to securely ingest credit/debit card streams, facilitating future geographical rollout out of Kenya into Uganda, Tanzania, and West Africa.
- **Platform Escrow Protection:** Secure ledger architecture holding transaction funds until fulfillment tracking status resolves to DELIVERED, closing the historical trust gap in online commerce.

D. The AI Conversational Core ('The Bestie')

- **The 'AI-Bestie' Natural Language Core:** Contextual interface wrapper built on Gemini 1.5 Flash via Vertex AI. Configured via specialized system instructions to communicate fluently in a blend of English, Swahili, and Sheng using street-smart local phrasing.

- **Intelligent Intent Tool-Calling:** Converts spoken or typed phrases (e.g., '*Niko Eldoret, nishow a cheap shoe plug in CBD*') into clean Firestore database queries, serving dynamic product carousels right inside the chat bubble.
- **Multimodal Visual Match Scanning:** Vision model ingestion allowing users to snap pictures or upload screenshots of items. The engine analyzes the profile, cross-references it with live vendor databases, and tags the exact physical stall location and stock levels.